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Unlocking Multimodal Mathematical Reasoning via Process Reward Model

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Abstract

Process Reward Models (PRMs) have shown promise in enhancing the mathematical reasoning capabilities of Large Language Models (LLMs) through Test-Time Scaling (TTS). However, their integration into multimodal reasoning remains largely unexplored. In this work, we take the first step toward unlocking the potential of PRMs in multimodal mathematical reasoning. We identify three key challenges: (i) the scarcity of high-quality reasoning data constrains the capabilities of foundation Multimodal Large Language Models (MLLMs), which imposes further limitations on the upper bounds of TTS and reinforcement learning (RL); (ii) a lack of automated methods for process labeling within multimodal contexts persists; (iii) the employment of process rewards in unimodal RL faces issues like reward hacking, which may extend to multimodal scenarios. To address these issues, we introduce **URSA**, a three-stage **U**nfolding multimodal p**R**ocess-**S**upervision **A**ided training framework. We first construct MMathCoT-1M, a high-quality large-scale multimodal Chain-of-Thought (CoT) reasoning dataset, to build a *stronger math reasoning foundation MLLM*, URSA-8B. Subsequently, we go through an automatic process to synthesize process supervision data, which emphasizes both logical correctness and perceptual consistency. We introduce DualMath-1.1M to facilitate the training of URSA-8B-RM. Finally, we propose **P**rocess-**S**upervised **G**roup-**R**elative-**P**olicy-**O**ptimization (**PS-GRPO**), pioneering a *multimodal PRM-aided online RL method* that outperforms vanilla GRPO. With PS-GRPO application, URSA-8B-PS-GRPO outperforms Gemma3-12B and GPT-4o by 8.4% and 2.7% on average across 6 benchmarks. Code, data and checkpoint can be found at https://github.com/URSA-MATH.

1 Introduction

Following the substantial progress of Large Language Models (LLMs) in math reasoning [1–8], the math reasoning capabilities of Multimodal Large Language Models (MLLMs) have increasingly garnered attention [9–13]. Previous work has typically focused on aspects such as math reasoning data curation [14–18], training math-intensive vision encoders [19, 20], enhancing vision-language alignment [11, 21], or the application of post-training techniques [22–24, 13]. Given the success of Process Reward Models (PRMs) in improving LLM reasoning through methods like Test-Time Scaling (TTS) [25, 26] and Reinforcement Fine-Tuning (ReFT) [27, 28], the application of PRMs to multimodal reasoning remains unexplored.

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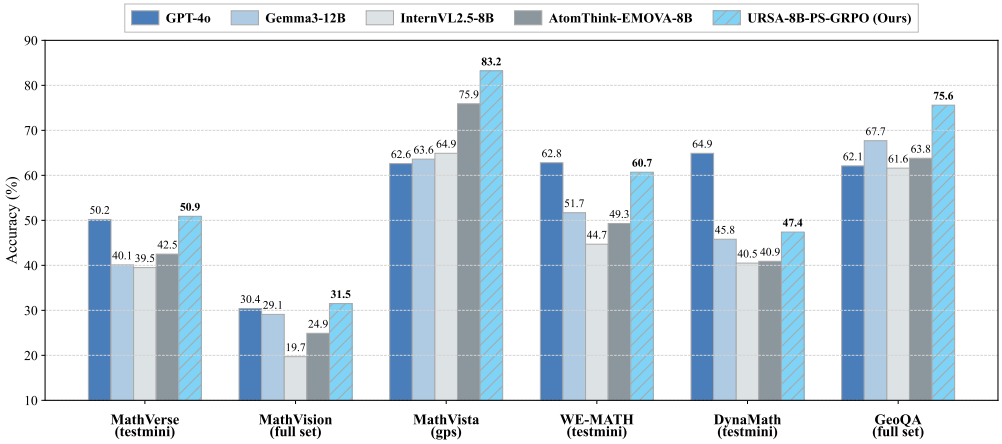

Figure 1: Performance comparison with leading open-source MLLMs and GPT-4o.

In this work, we take the first step toward integrating PRMs into multimodal math reasoning. We identify three key challenges: (i) Since both TTS and RL are heavily influenced by the strength of foundation models [29, 25], the limited availability of large-scale, high-quality reasoning data constrains the upper bounds of current MLLMs and weakens the effectiveness of PRM integration; (ii) There hasn't yet been adequate automated process labeling techniques merged within multimodal contexts, where both logical validity and perceptual consistency should be emphasized [30–32]. (iii) While PRMs can be effectively used in TTS, applying them directly in online RL introduces risks such as reward hacking and length bias in rewarding [33, 34].

To address these challenges, we propose the **URSA** framework, a three-stage **U**nfolding multimodal p**R**ocess-**S**upervision **A**ided training pipeline that supports both the construction and application of multimodal PRMs. In Stage I, we curate **MMathCoT-1M**, a large-scale, high-quality multimodal Chain-of-Thought dataset synthesized from 1.43 million open-source examples, which enhances the foundation model's reasoning capabilities through targeted instruction tuning. In Stage II, we construct **DualMath-1.1M** via a dual-view process supervised data synthesis strategy which combines a binary error locating engine and a misinterpretation insertion engine. It provides complementary signals for logical validity and visual grounding, and is used to train a process reward model. In Stage III, we analyze the limitations of scalar process reward modeling in online RL and propose **Process Supervision-GRPO (PS-GRPO)**, which mitigates reward hacking and PRM's length bias in rewarding by implicitly penalizing process-level inconsistencies during policy optimization.

Results on 6 multimodal reasoning benchmarks show that our PRM improves Best-of-N verification, surpassing self-consistency and outcome-based baselines. When used in PS-GRPO, the resulting model achieves state-of-the-art performance among open-source MLLMs of similar size. Our contributions are as follows:

- We release two large-scale open-source datasets, MMathCoT-1M and DualMath-1.1M, to address the scarcity of high-quality multimodal CoT reasoning and process supervision data.
- We propose PS-GRPO, an online reinforcement learning algorithm that incorporates multimodal PRMs by comparing the relative quality of rollouts, rather than relying on scalar reward modeling. It effectively mitigates PRM's reward hacking and length bias in rewarding.
- Experimental results show that our reward model improves both test-time verification and online training. With PS-GRPO application (Figure 1), URSA-8B-PS-GRPO outperforms Gemma3-12B and GPT-4o by 8.4% and 2.7% on average across 6 benchmarks.

2 Stage I: Math-Intensive Alignment and Instruction Tuning

2.1 Collection of Vision-Language Alignment Data

We employ a LLaVA-like architecture and first collect vision-language alignment data directly from existing open-source datasets [35–38]. As demonstrated in Figure 2, we collect URSA-Alignment-860K from Multimath [23], MAVIS [19] and Geo170K [18]. We then filter out samples with overly

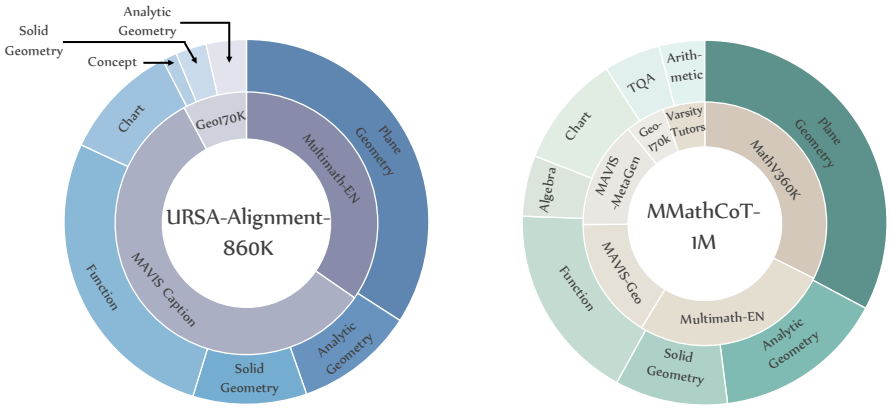

Figure 2: Statistics of URSA-Alignment-860K and MMathCoT-1M.

verbose captions, to form an 860K math-intensive alignment dataset. Following the engineering practices of previous work, we only train the MLP projector in the alignment step.

2.2 CoT Reasoning Data Synthesis

For a powerful foundation building, we collect 1.43M samples from existing math reasoning datasets to support the construction of large-scale CoT reasoning data. As shown in Figure 2, data is sourced from MathV360K [15], Multimath [23], MAVIS [19], Geo170K [18] and VarsityTutors [11]. Based on the type of solution, we categorize the data into *answer-only*, *analysis-formatted*, and *CoT-formatted*. We adopt different synthesis strategies for them to curate high-quality CoT reasoning trajectories. We utilize Gemini-1.5-Flash-002 (refer to $\mathcal{G}$ below) as a cost-effective tool for data curation, avoiding expensive large-scale manual annotation.

CoT Expansion. For *answer-only* data $\mathcal{D}_1 = \{(x_i, y_i)\}_{i=1}^{N_1}$, such as MathV360K [15], each sample contains a question x_i and a ground-truth answer y_i. This type of data is heavily used in previous works for fast thinking reasoning mode [15, 11, 16]. However, answer-only training restricts the model from fully capturing the problem-solving process. It may lead to memory-based reasoning, hindering the model's ability to directly provide answers to more complex reasoning problems [39]. We expand certain scale CoT reasoning trajectories for this category of data. Given a expansion prompt $\mathcal{P}_{\mathcal{C}}$, we provide x_i and y_i, then prompt $\mathcal{G}$ to output the reasoning trajectory leading to the answer y_i, yielding the expanded solutions $\mathcal{S}_{Ao} = \mathcal{G}(\mathcal{P}_{\mathcal{C}}; \{x_i, y_i\}_{i=1}^{N_1})$.

Rewriting. This strategy is designed for *analysis-formatted* samples, denoted as $\mathcal{D}_2 = \{(x_i, y_i, a_i)\}_{i=1}^{N_2}$. This includes datasets like MAVIS-Geo, MAVIS-MetaGen [19], VarsityTutors [11], and Geo170K-QA [40]. Each sample contains a question x_i, an answer y_i, and textual analysis a_i. While this type of data provides walkthroughs, it often suffers from two issues: (i) It lacks strict step-by-step logic, exhibiting jumps in language or reasoning. (ii) A significant portion of the answers are relatively brief and cannot provide rich rationale. Given a rewriting prompt $\mathcal{P}_{\mathcal{R}}$, we utilize $\mathcal{G}$ to transcribe these solutions, thereby enhancing their step-by-step reasoning trajectories and linguistic diversity, resulting in the rewritten set $\mathcal{S}_{An} = \mathcal{G}(\mathcal{P}_{\mathcal{R}}; \{x_i, y_i, a_i\}_{i=1}^{N_2})$.

Format Unification. This strategy is used for *CoT-formatted* data, primarily sourced from Multimath-EN-300K [23], which is collected from K-12 textbooks and contains mathematical language and symbolic-style reasoning solutions. This portion of the data, $\mathcal{D}_3 = \{(x_i, y_i, c_i)\}_{i=1}^{N_3}$, consists of a question x_i, an answer y_i, and a solution c_i. We unify the format through natural language stylization using a prompt $\mathcal{P}_{\mathcal{F}}$ with $\mathcal{G}$, producing the unified set $\mathcal{S}_C = \mathcal{G}(\mathcal{P}_{\mathcal{F}}; \{x_i, y_i, c_i\}_{i=1}^{N_3})$.

MMathCoT-1M. Finally, we filter out instances where: (i) Correctness is violated: the generated content altered the original answer, or (ii) Consistency is problematic: the solution includes text that questions the original answer or makes new assumptions to force the given answer. This process yields MMathCoT-1M. The complete prompt designs can be found in Appendix G.